MODULE 7

Purification & Conditioning

Separation · drying · PSA · membranes · de-oxo · compression · liquefaction

7.1 Why purify?

Raw hydrogen from any process carries impurities: water and traces of O_2 (electrolysis), or CO_2, CO, CH_4, N_2 and H_2O (reforming/gasification). The end use sets the spec – for example, PEM fuel cells are very sensitive to CO and sulphur, so vehicle hydrogen must meet **ISO 14687**, while a refinery hydrotreater tolerates more impurities.

Technology	Principle	Removes	Where used
Gas–liquid separation	Gravity + demister	Liquid water / electrolyte droplets	Directly after every electrolyser
Drying – TSA	Adsorption on molecular sieve; heat regeneration	Moisture	Electrolysis plants; pipeline gas
Drying – refrigeration	Cool to condense water	Bulk moisture (pre-step)	Ahead of TSA / compressors
PSA	Impurities adsorb at high pressure, release at low pressure	CO_2, CO, CH_4, N_2, H_2O	SMR/ATR/gasification; by-product H_2 recovery
Membrane separation	H_2 permeates faster through polymer/metal membrane	Bulk separation; Pd membranes give very high purity	Refinery off-gas recovery, small units
Deoxidation (de-oxo)	Catalytic $2H_2 + O_2 \rightarrow 2H_2O$	Trace O_2	Electrolysis plants
Cryogenic separation	Partial condensation at low temperature	Hydrocarbons, N_2	Large refinery / petrochemical streams

7.2 Pressure Swing Adsorption – how a cycle works



Several beds run out of phase so product flow is continuous. Hydrogen is barely adsorbed, so it passes through; the tail gas (impurities + some H_2) is burned in the SMR furnace. Trade-off: higher purity ↔ lower hydrogen recovery.

7.3 Compression

- **Reciprocating piston** – high flow and pressure; needs careful sealing (lubricated or oil-free designs).
- **Diaphragm** – metal diaphragm isolates gas from oil; standard for high-purity, high-pressure duty such as refuelling stations.
- **Ionic liquid** – liquid piston; few moving parts.
- **Electrochemical** – proton-conducting cell pumps H_2; no moving parts, small scale.
- Electrolysers can deliver pressurised gas directly (up to ~30 bar typical, higher possible), but higher stack pressure increases gas crossover and losses [R1].

7.4 Liquefaction

Hydrogen is cooled below –253 °C in stages (pre-cooling with liquid nitrogen or mixed refrigerant, then a hydrogen Claude/Brayton cycle with expansion turbines). An **ortho-to-para conversion** catalyst is required because the natural ortho→para conversion releases heat that would otherwise boil off stored liquid. Liquefaction reduces volume ~870× vs atmospheric gas [R1] but consumes a large share of the hydrogen's energy content, so it is used mainly for long-distance or high-volume delivery.

7.5 Storage

Compressed, liquid, underground, hydride and chemical-carrier storage are covered in Module 8.

Quiz & revision – Module 7

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Which step removes trace oxygen from electrolytic H2?
a) PSA **b)** De-oxo **c)** Membrane **d)** Cyclone

Q2. In PSA, what happens to impurities?
a) Pass through **b)** Adsorb at high pressure **c)** React to water **d)** Freeze

Q3. Why is ortho–para conversion needed in liquefaction?
a) To increase density **b)** Conversion heat would boil off stored LH2 **c)** To remove CO **d)** To raise pressure

Q4. Fuel-cell vehicle hydrogen quality standard:
a) ISO 9001 **b)** ISO 14687 **c)** IEC 61850 **d)** ASME VIII

Q5. Main drawback of higher PSA purity:
a) Lower H2 recovery **b)** More CO2 **c)** Higher water use **d)** No drawback

Answer key
Q1 – b: ; **Q2 – b**: then released at low pressure; **Q3 – b**: ; **Q4 – b**: ; **Q5 – a**: